



Activity Diagram

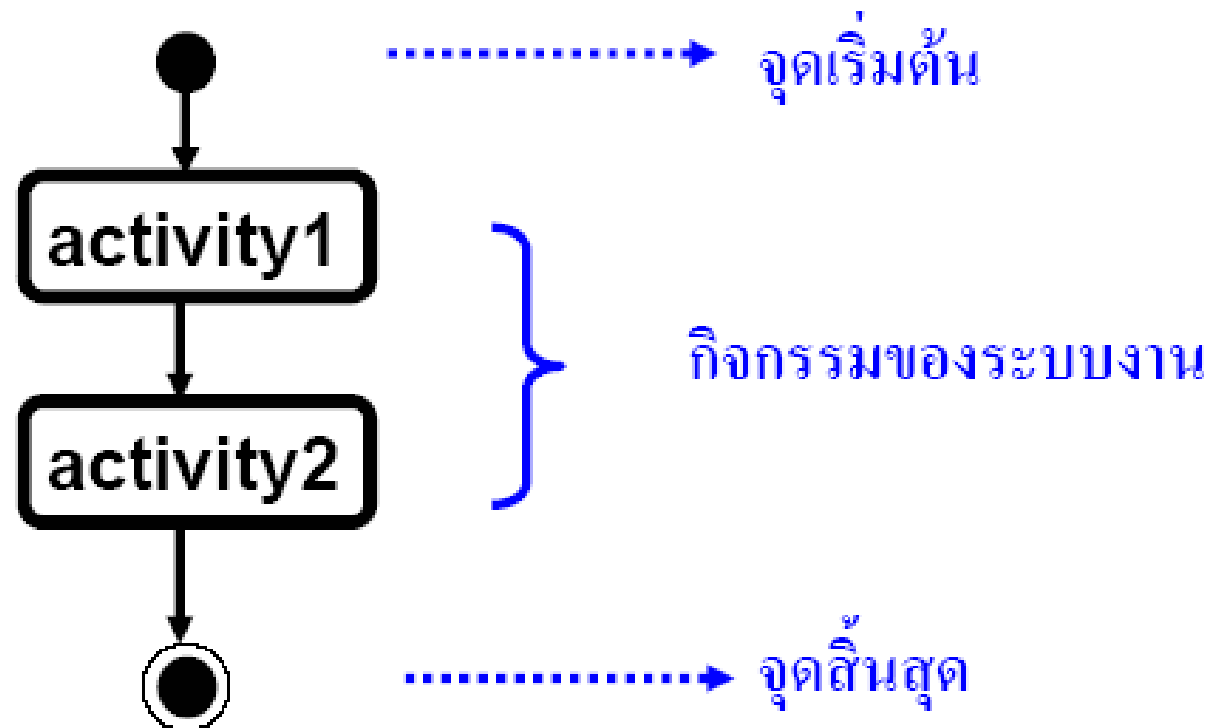
Activity Diagrams

The Activity Diagram follows the user through a procedure, noting the decisions made and tasks performed. The procedure may explain a single use case, or only part of a use case, or even many use case tied together to create a workflow.

- Use: Understanding Work-Flow
- Use: Analyzing Use-Case
- No: Analyzing Object Collaboration
 - Use Sequence or Collaboration Diagrams
- No: Analyzing Object Behavior
 - Use State Diagram

Notation

สัญลักษณ์



Start Point

- A start point is modeled with a filled in circle, using the same notation that UML State Chart diagrams use. Every UML Activity Diagram should have a starting point, and placing it in the top-left corner reflects the way that people in Western cultures begin reading.

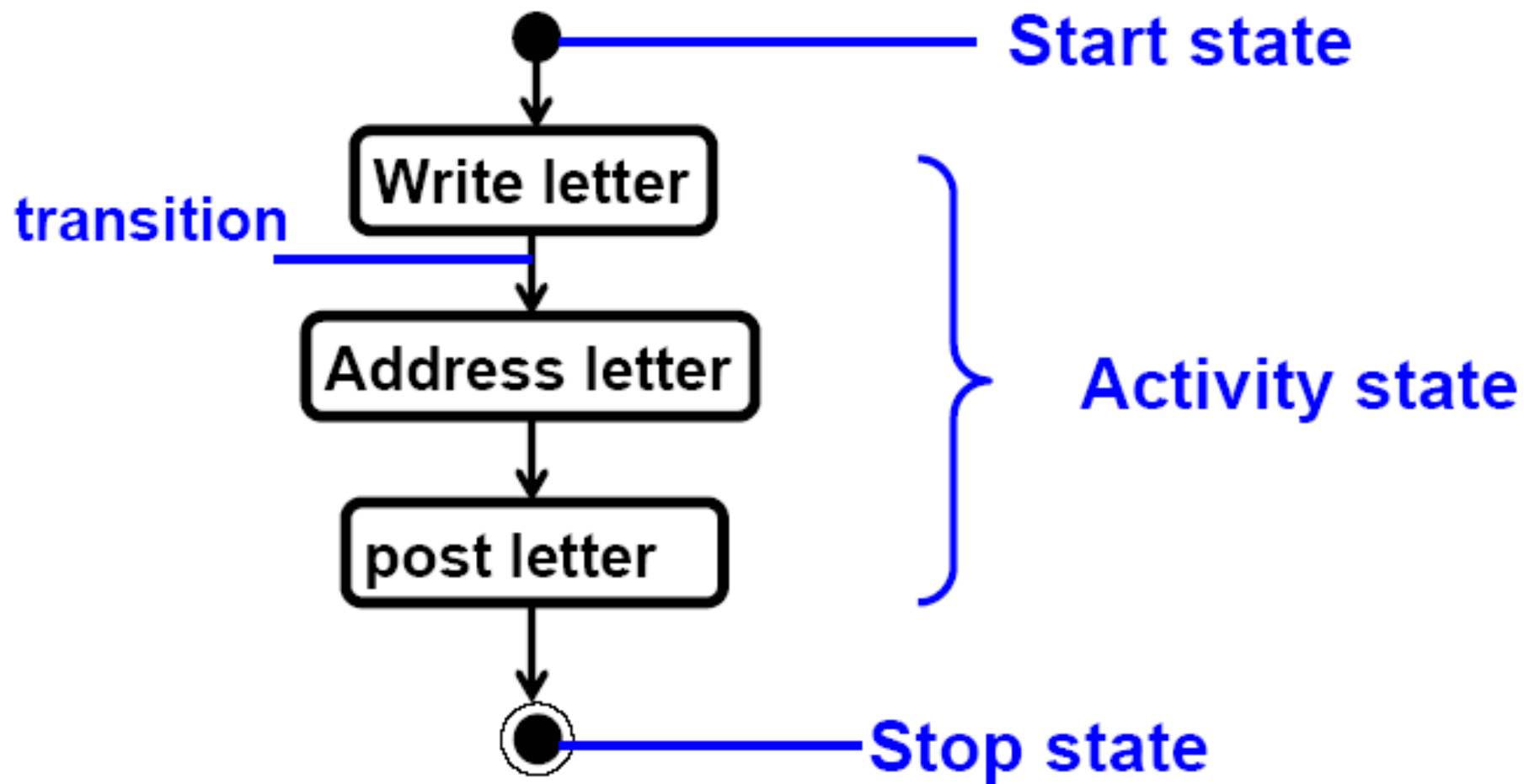
End Point

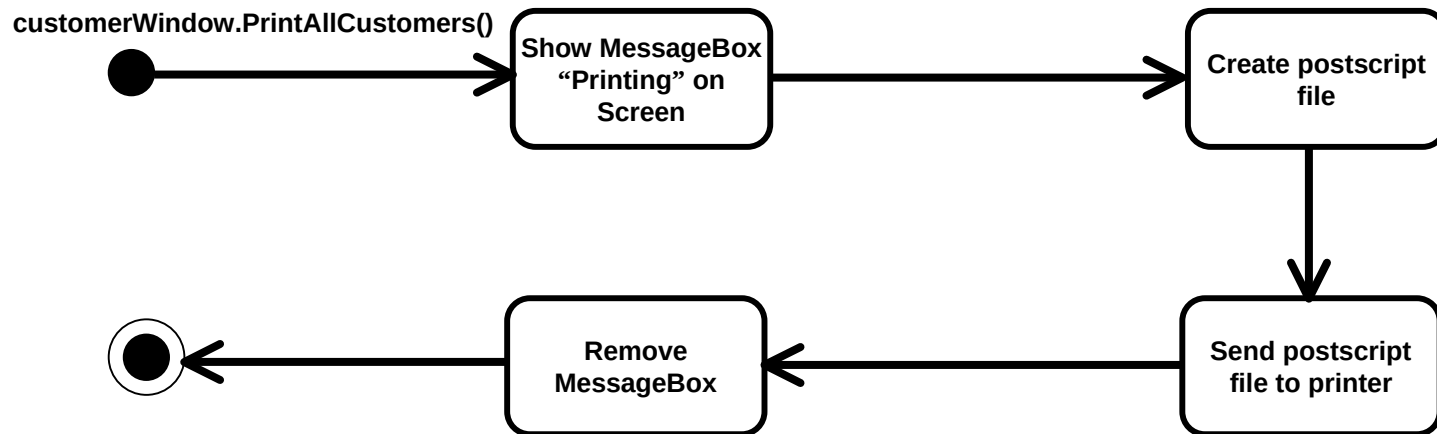
- An ending point is modeled with a filled in circle with a border around it, using the same notation that UML State Chart diagrams use. Fowler and Scott's (1999) style is to make ending points optional, some times an activity is simply a dead end, but if this is the case then there is no harm in indicating that the only transition is to an ending point. That way when someone else reads your diagram they know that you have considered how to exit these activities.

State, Transitions, Guard Condition

- *Activity states*, which represent the performance of a step within the workflow.
- *Transitions* that show what activity state follows after another. This type of transition can be referred to as a completion transition. It differs from a transition in that it does not require an explicit trigger event; it is triggered by the completion of the activity that the activity state represents.
- *Guard condition* can be assigned to a transition to restrict its use. Place the condition within square brackets somewhere near the transition arrow.

เช่น





When someone calls the `PrintAllCustomer` operation (in the `CustomerWindow` class), the actions start. The first action is to show a message box on the screen; the second action is to create a postscript file; the third action is to send the postscript to the printer; and the fourth action is to remove the message box from the screen. The transitions are automatic; they occur as soon as the action in the source state is performed.



ใน activity diagram ยังมีลักษณะ

- Swimlanes

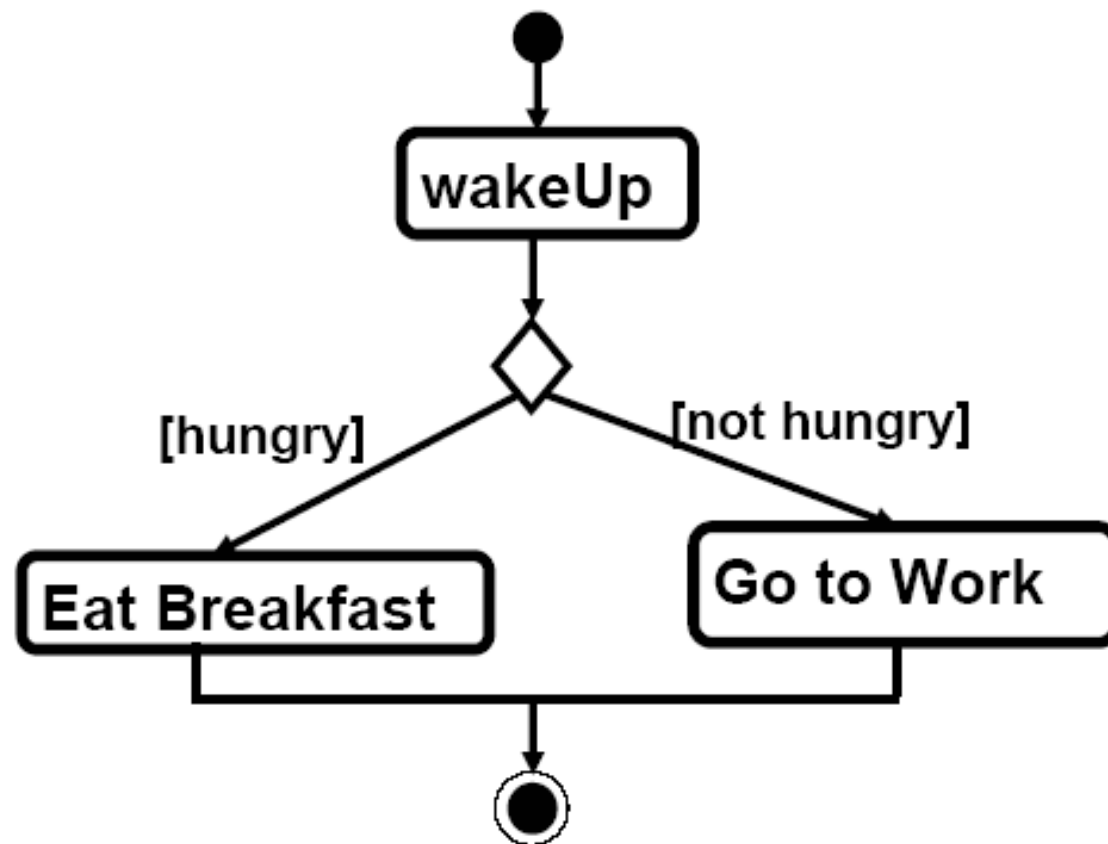
- Decision

- Forks and joins

Decision

มีทางเลือกให้ตัดสินใจ

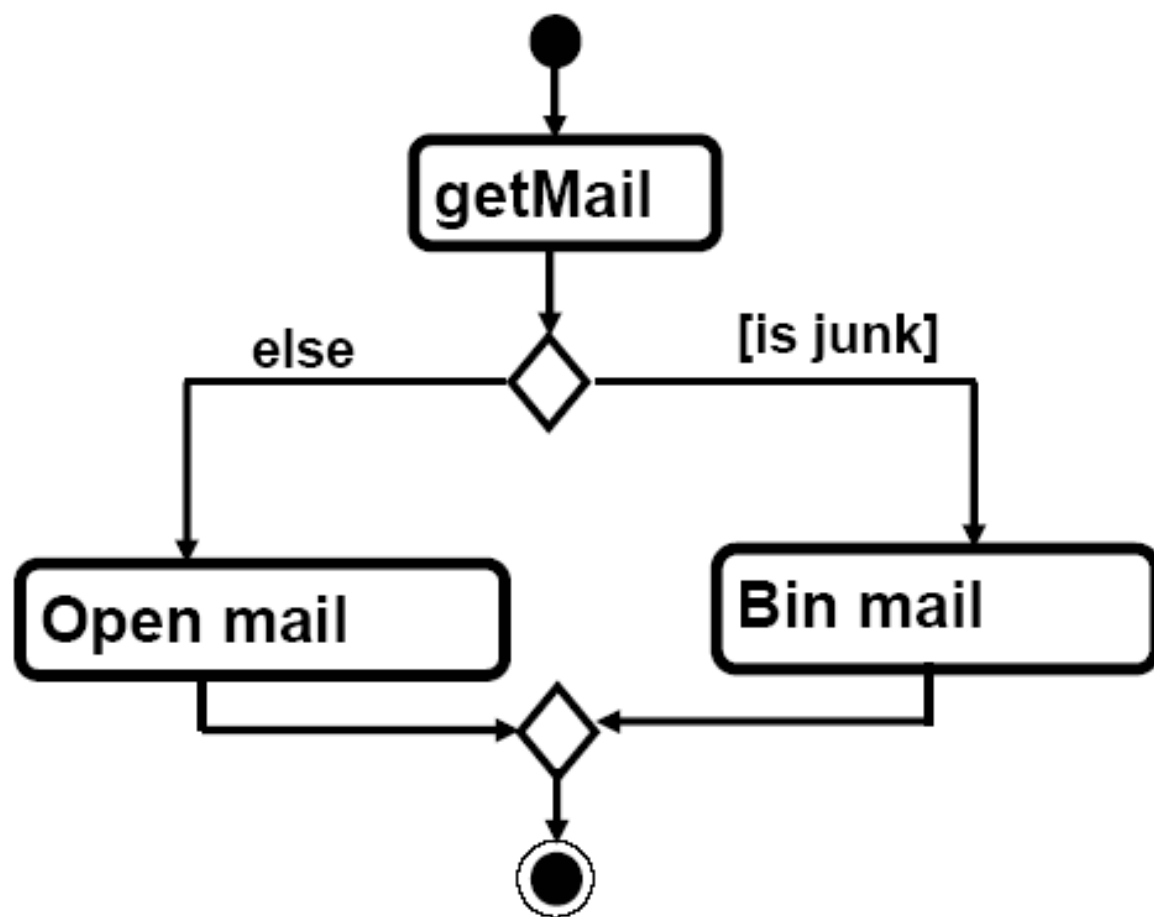
สัญลักษณ์
เช่น



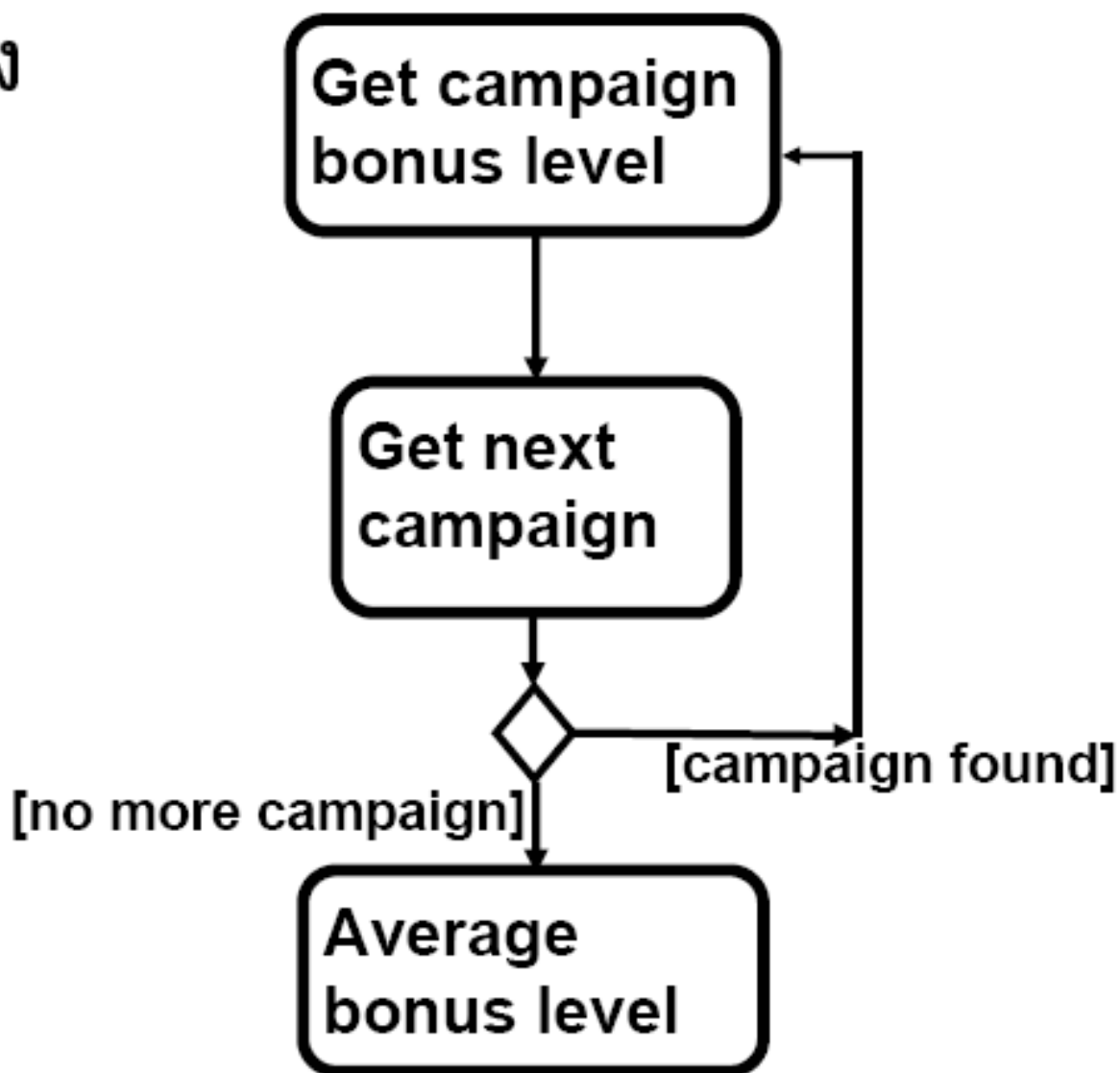
Decision

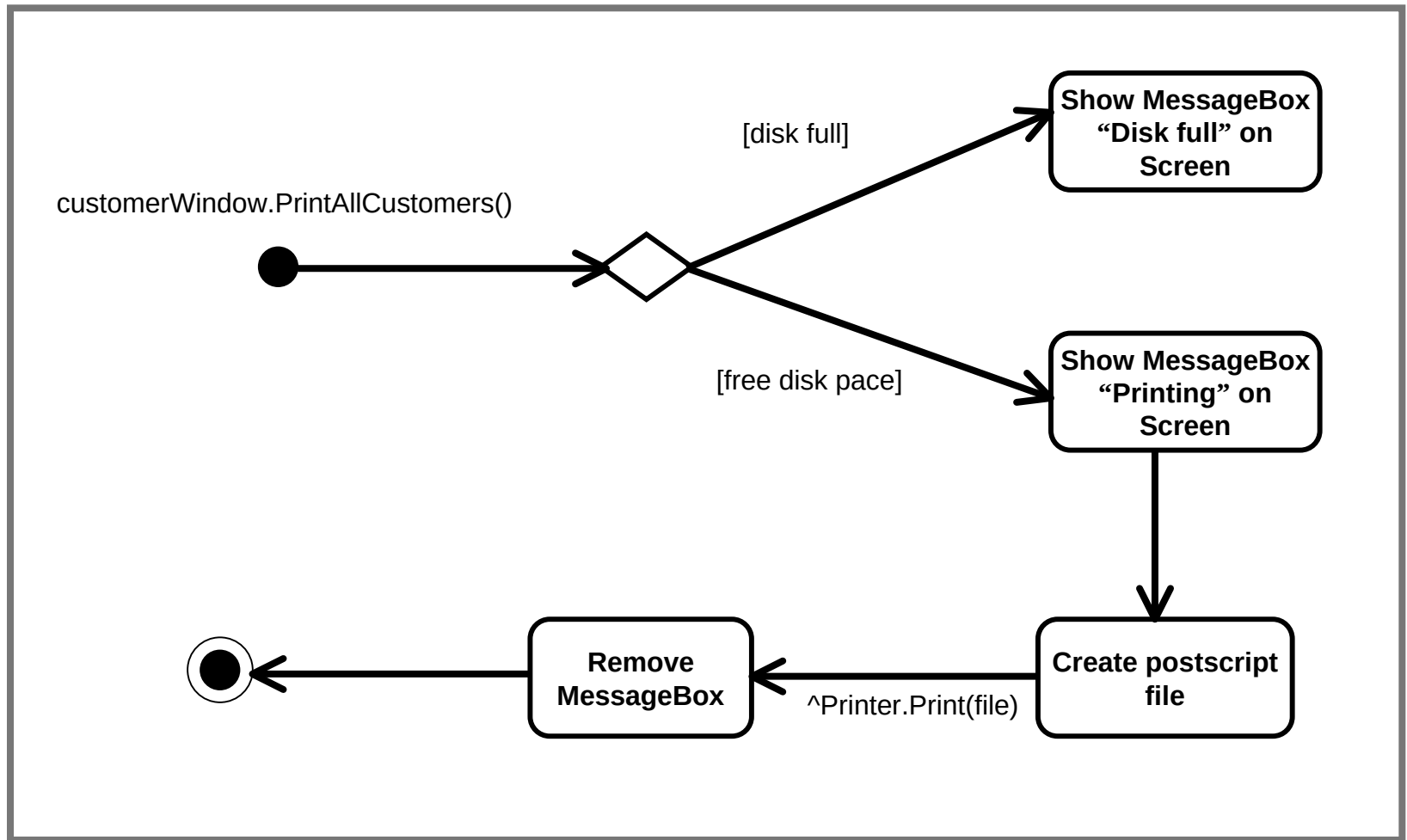
- In Figure you see that there is no label on the decision point, unlike traditional flowcharts which would include text describing the actual decision being made, you need to imply that the decision concerns whether the person was enrolled in the university based on the activity that the decision point follows. The guards, depicted using the format *[description]*, on the transitions leaving the decision point also help to describe the decision point.

ตัวอย่าง



ตัวอย่าง

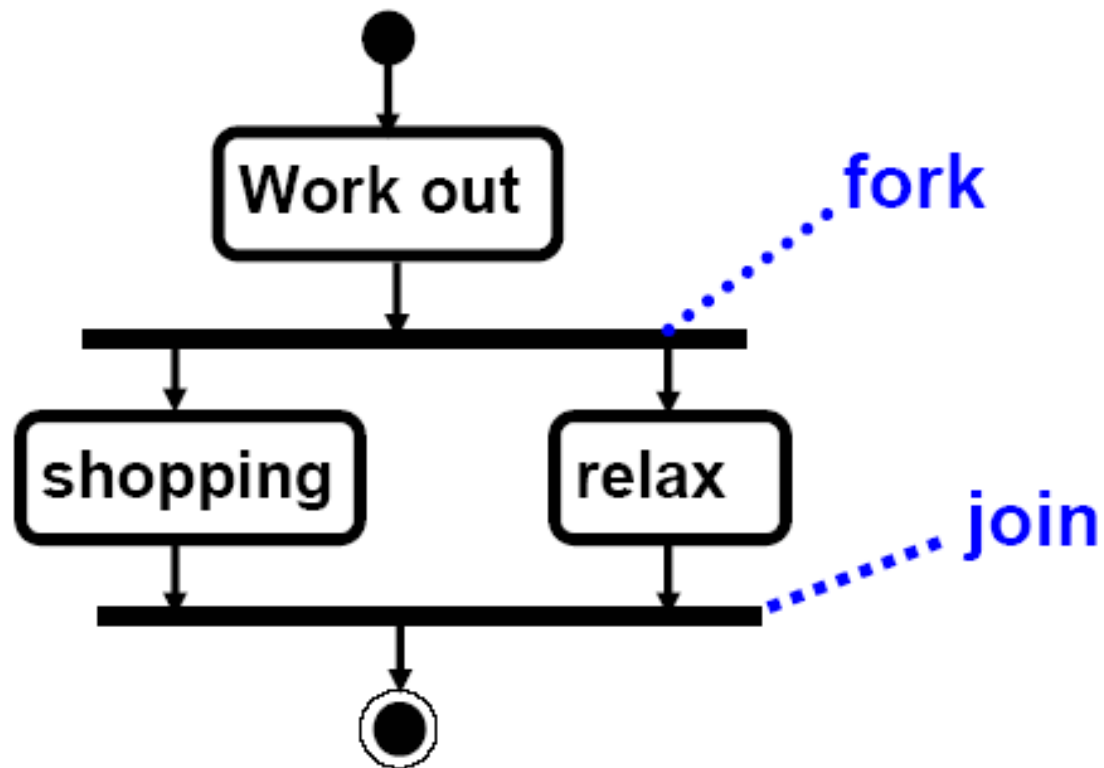




If the disk is full, MessageBox “Disk full” is shown. Otherwise MessageBox “Printing” is show.

Forks and joins

งานหลายงานที่มีการทำงานไปพร้อมๆ กัน
สัญลักษณ์ ใช้เส้นตรงแนวนอนบนเส้นหนา
ตัวอย่าง



Forks and Join

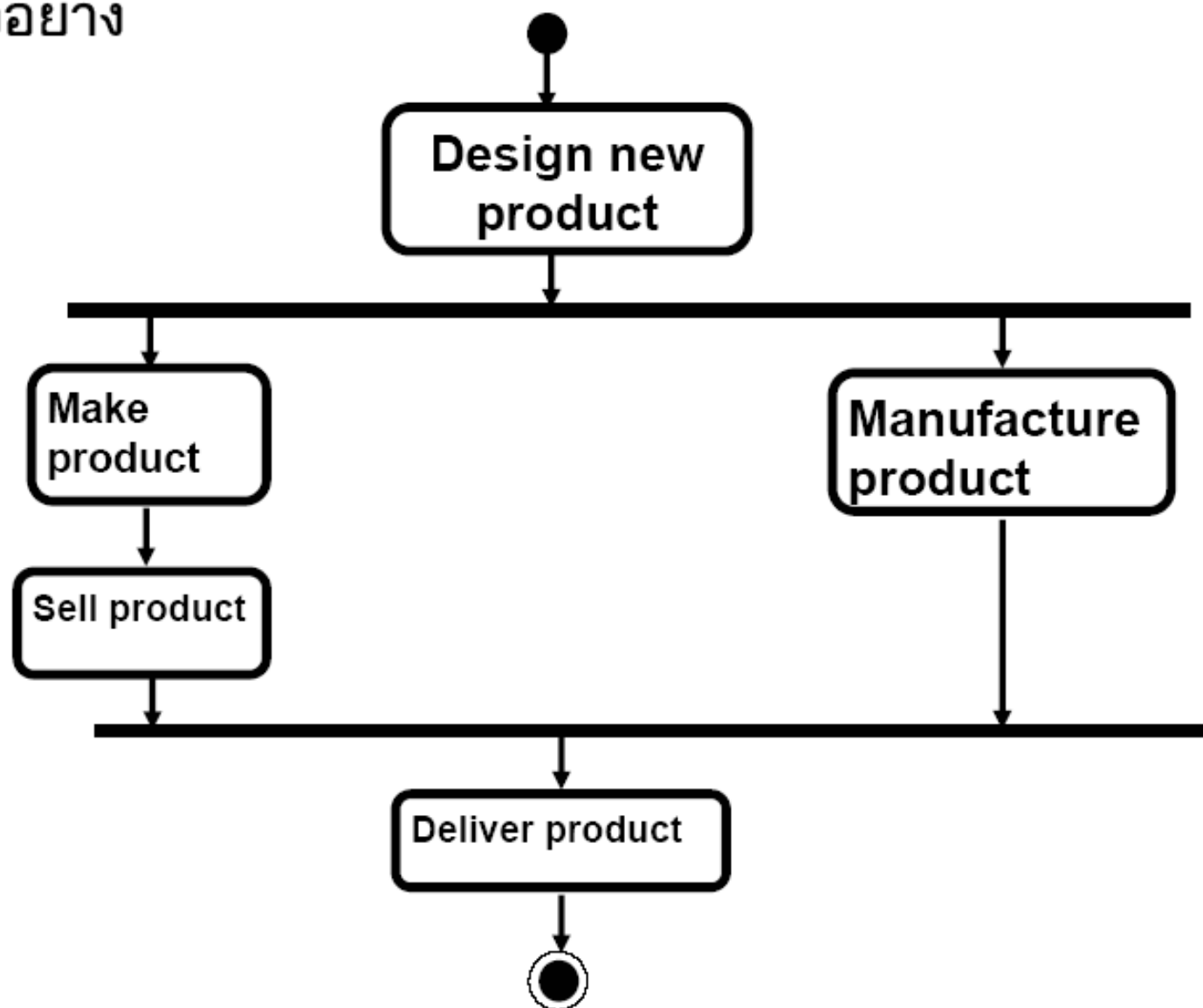
Forks Have One Entry Transition

- When you find that you want to have several transitions into the same fork you should first merge them by having them enter a single diamond and then have a single transition from the diamond into the fork. However, this situation is also a good indication that you've either missed an activity, likely where the merge occurs, or that you really don't have parallel activities at this point.

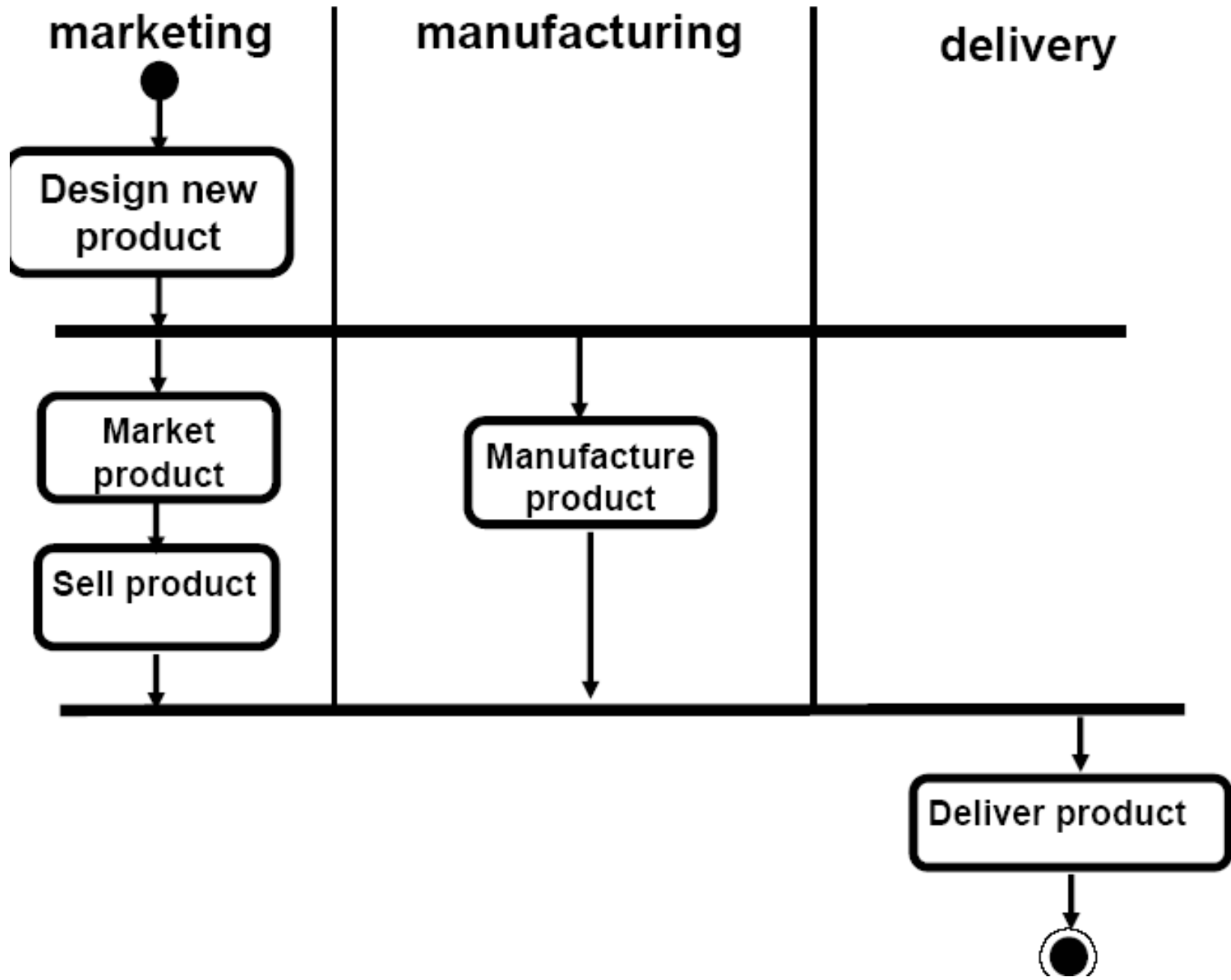
Joins Have One Exit Transition

- The desire to have several exit transitions is a good indication that you still have parallel activities occurring; therefore move your join further along in the overall process.

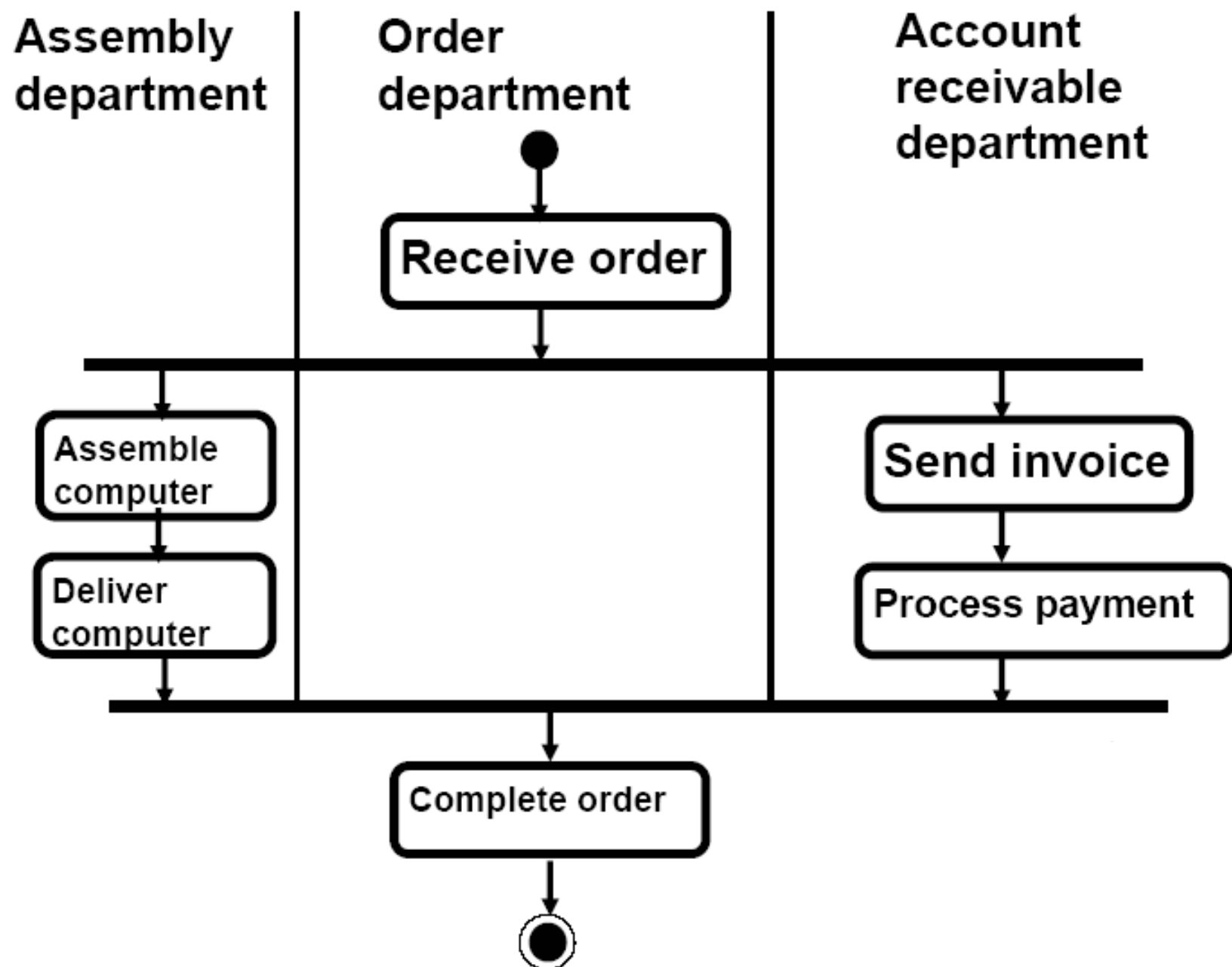
ตัวอย่าง



ตัวอย่าง activity diagram for a company assembly company



ตัวอย่าง activity diagram for a computer assembly company



Swimlanes

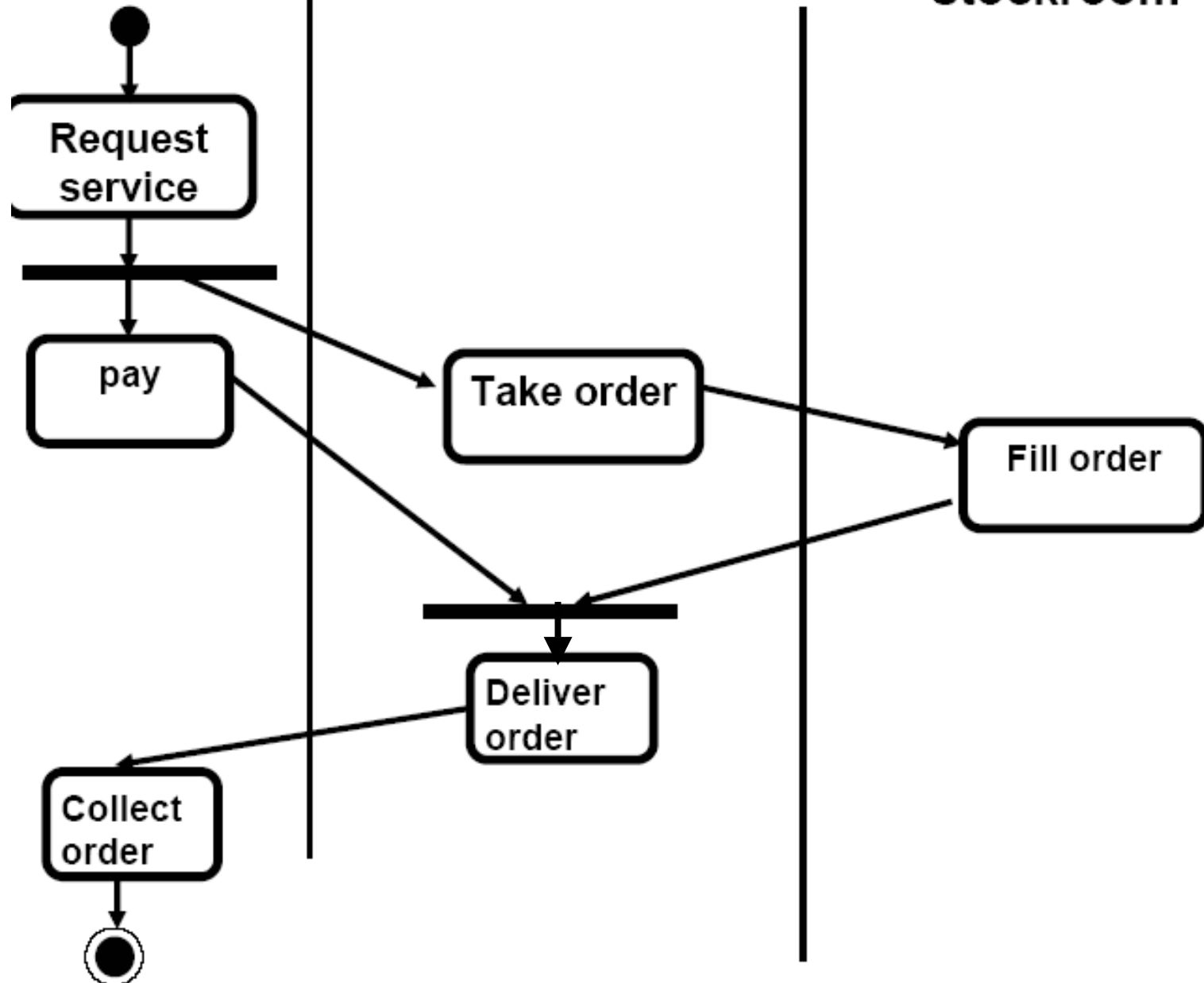
เป็นการแบ่งกลุ่ม activity เป็นเลน เหมือนสระว่ายน้ำ
แบ่งเป็นช่องในแนวดิ่ง และกำหนดแต่ละช่องด้วยชื่อ
object ไว้ด้านบน

การแบ่งเป็น Swimlanes ช่วยให้แยกแยะผู้รับผิดชอบแต่ละงานได้ว่า ใครควรจะเป็นคนทำงานในหมวดหมู่ใด
ตัวอย่างเช่น

customer

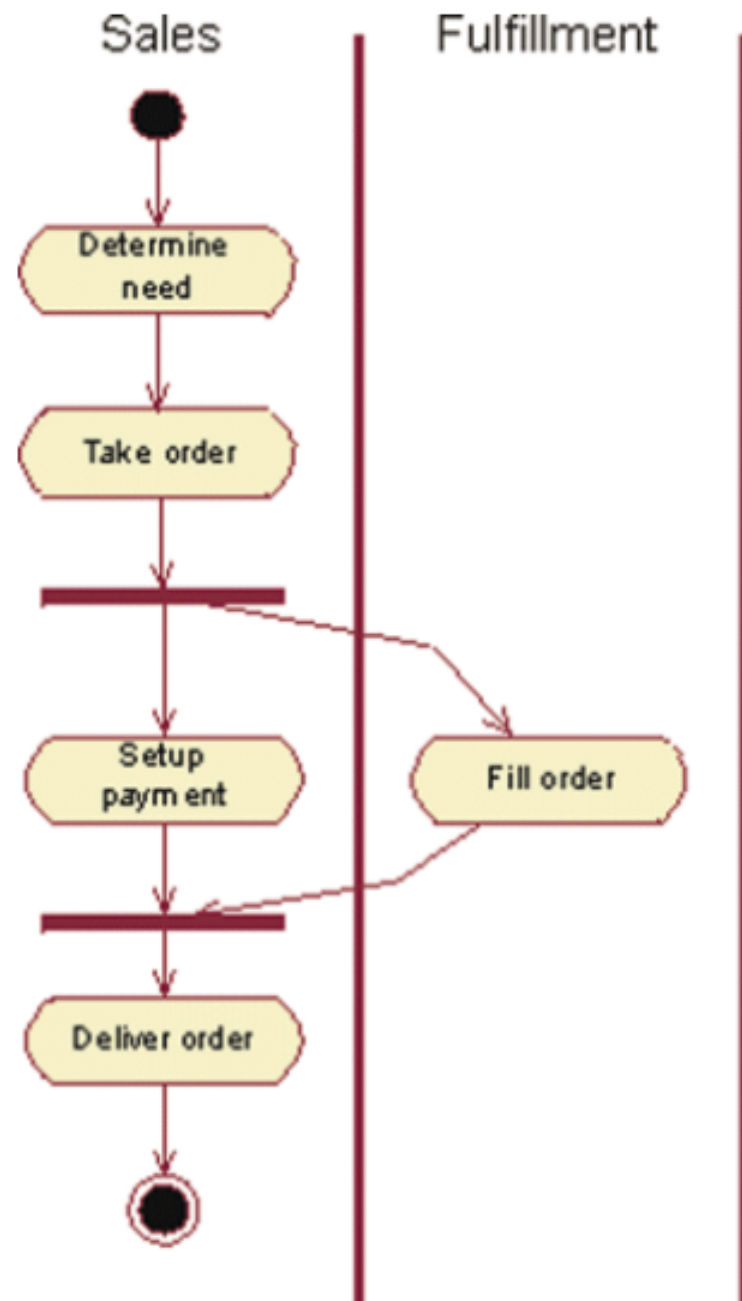
sales

stockroom



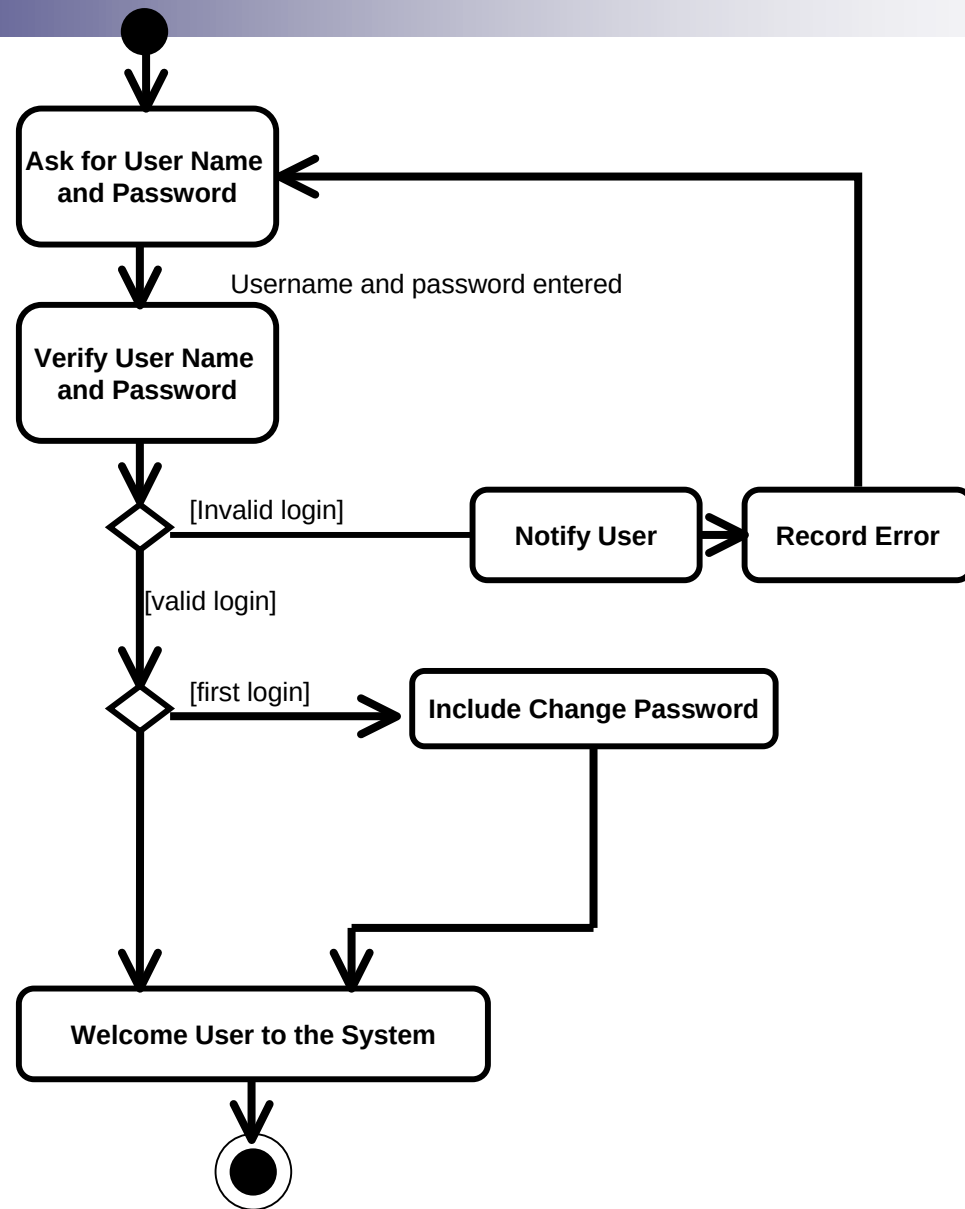
An Activity Diagram Illustrating the Workflow of a Business Use Case that Represents a (Generic) Sales Process.

In this example, the partitions represent departments in the organization.



Activity diagram

Login Use Case



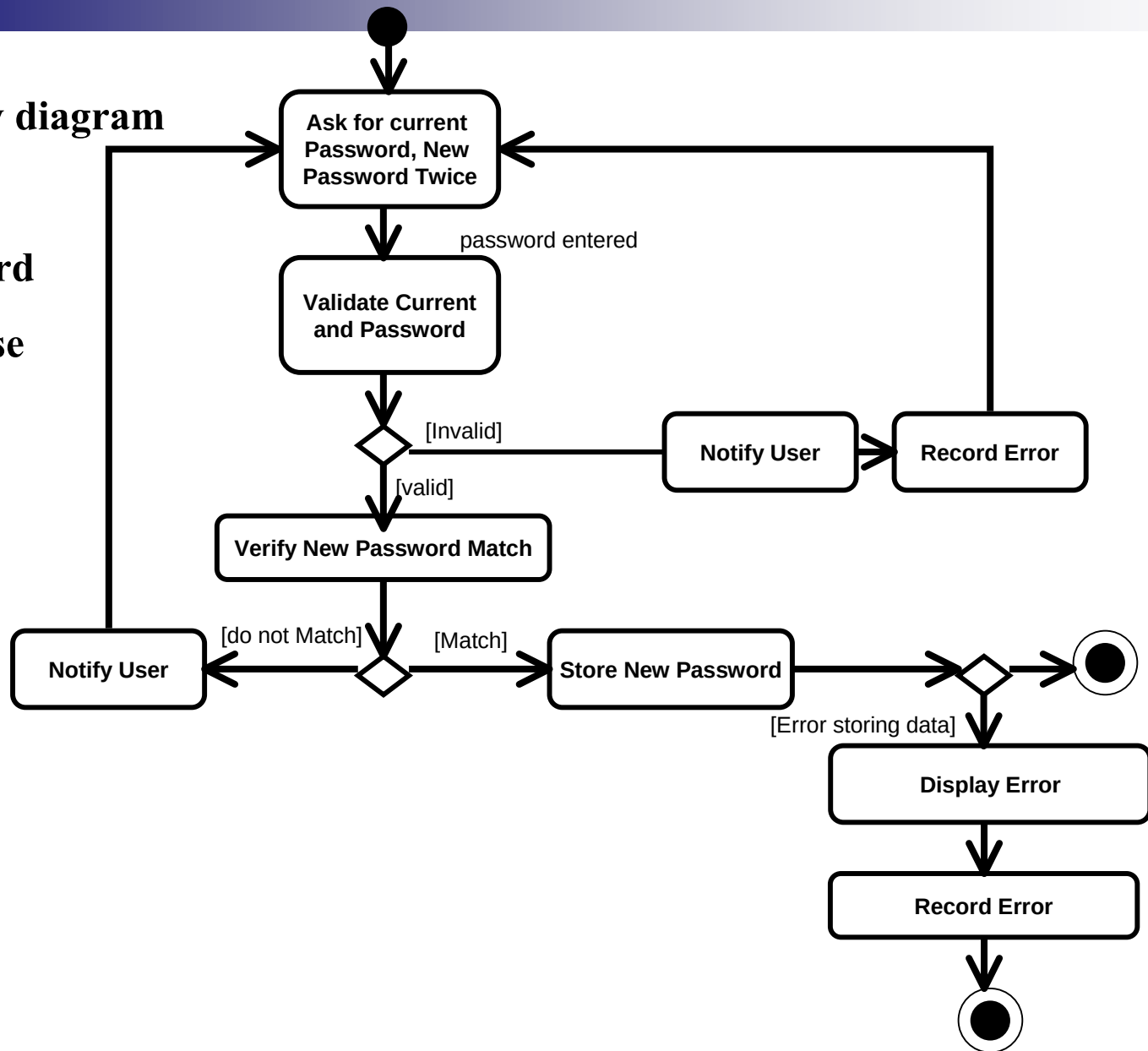
An activity diagram used to describe the interaction between the actor and the use case.

Activity diagram

Change

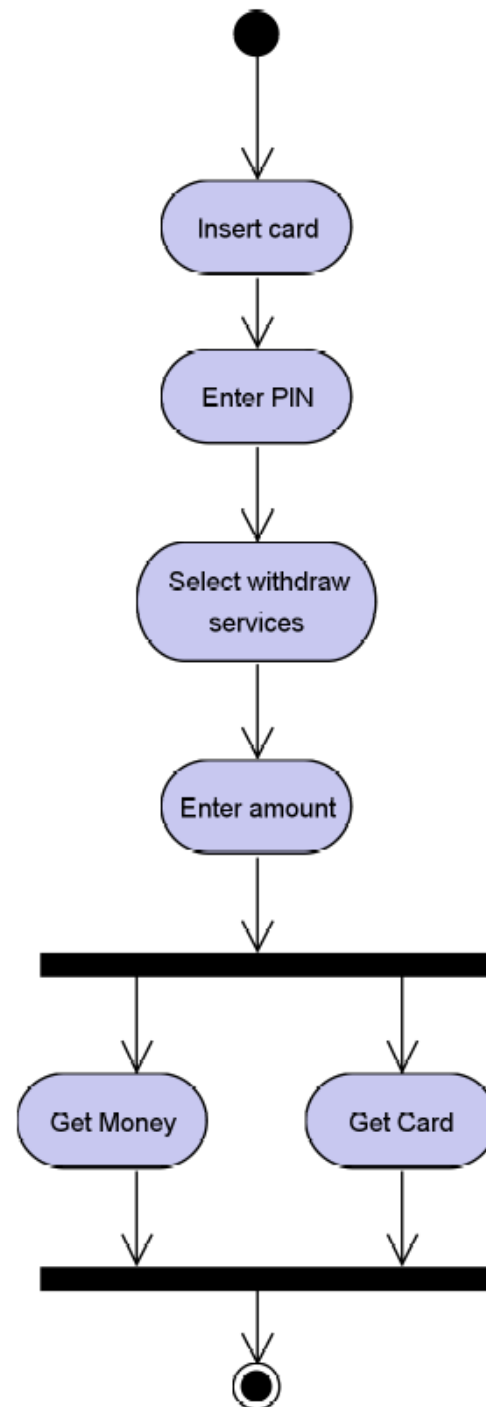
Password

Use Case



ATM Example System

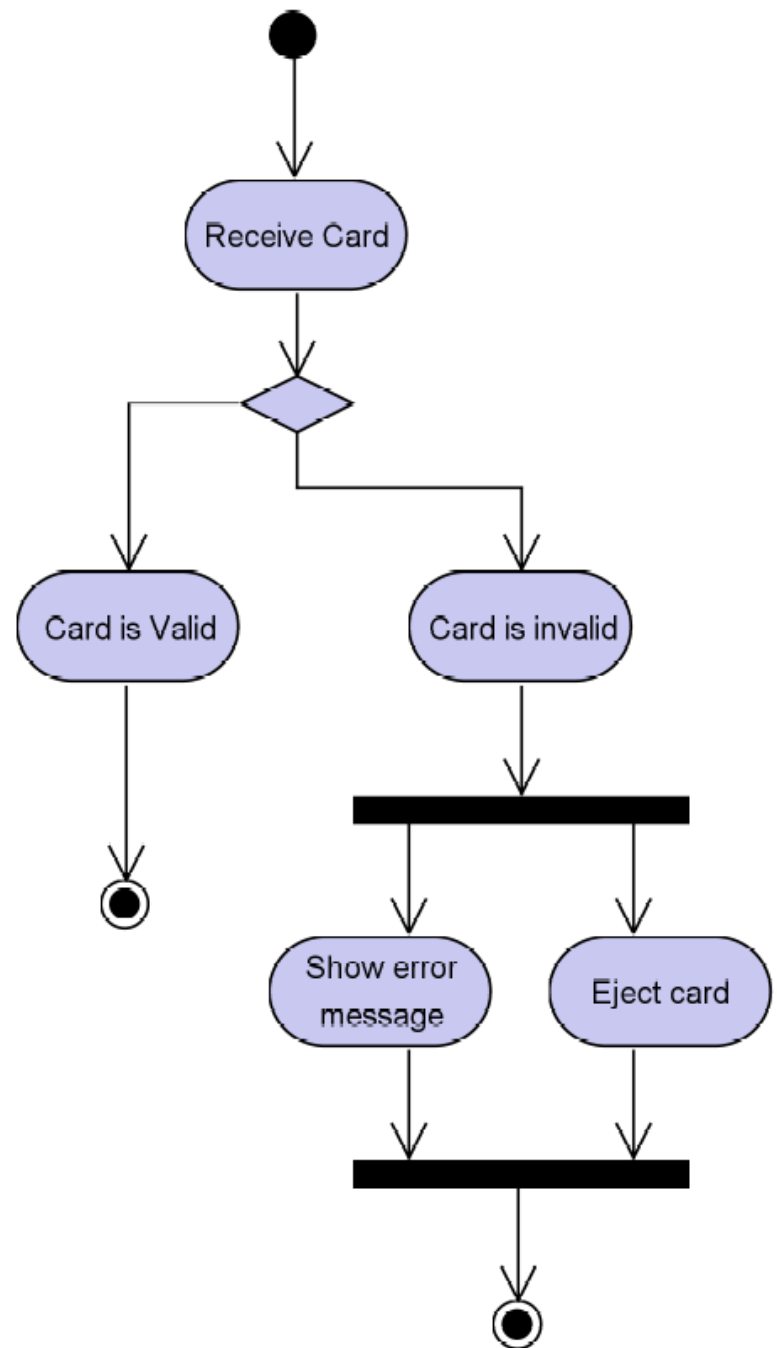
User Activity Diagram



ATM Example System

Card Controller

Activity Diagram



Withdraw money in the use case model of a ATM

